Assignment 2 - Git & GitHub Fundamentals
Report Template

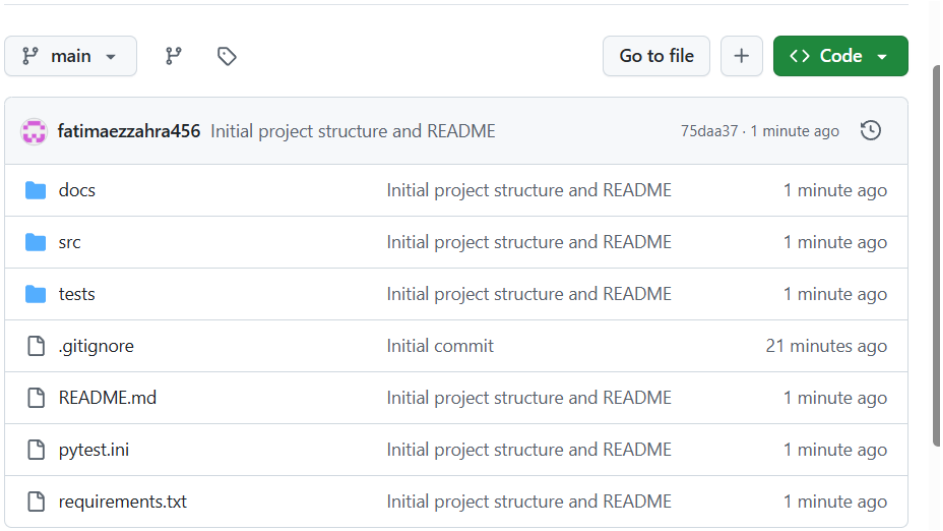
- ***Group Size:*** 3 members (3-member path)
- ***Chosen Theme:*** To-Do List
- ***Full Names:*** Fatima Ezzahra Ghezala (Member A); Hafsa El Mahfoudy (Member B); Chaimae Tahir (Member C)
- ***Student IDs:*** 163591, 158085**,** 120024
- ***Course Codes & Sections:*** CSC3324
- ***GitHub Usernames:*** fatimaezzahra456 (A); Hafsa-mhf (B); chaimaetahir (C)
- ***GitHub Repository URL:*** https://github.com/fatimaezzahra456/To-do-list-project
- ***GitHub Pages URL:*** https://fatimaezzahra456.github.io/To-do-list-project/

1. Project & Role Distribution

The group chose the **To-Do List** theme. The program logic is written in Python and tested with pytest. There is no database, backend or external API. A task is a small dictionary with an id, a title and a done flag. The group follows the **3-member path**, with three comparable feature areas:

- **Member A, Fatima Ezzahra Ghezala (fatimaezzahra456):** task management (add_task, complete_task) in src/tasks.py. Also created the repository, the ruleset, the CI workflow and the project page.
- **Member B, Hafsa El Mahfoudy (Hafsa-mhf):** filter and search (filter_by_status, search_tasks) in src/filters.py.
- **Member C, Chaimae Tahir (chaimaetahir):** statistics (completion_percentage, count_pending) in src/stats.py.

Member A created the public repository with the initial structure (src/, tests/, docs/, requirements.txt, pytest.ini, .gitignore) and a README.md listing the theme, the members table and the line “Project status: Development in progress.” Each member cloned the repository and worked from a local copy.



Repository structure on GitHub (public repository).

2. Branches, Commits & Contributions

Member A: Fatima Ezzahra Ghezala (fatimaezzahra456)

- **feature-tasks:** src/tasks.py and tests/test_tasks.py. Commits: 387efc2 “Add add_task and complete_task functions” and 9d806f5 “Add unit tests for task management”.
- **ci-workflow:** .github/workflows/tests.yml and tests/test_ci_demo.py. Commits: 69844f6 “Add CI workflow with demo test” (intentionally failing test) and 1afbd1c “Fix failing CI demo test”.
- **status-update-fatima** (merge conflict): 32fb214 “Update project status (Fatima)” and the conflict-resolution merge commit 97aab98 “Resolve README status conflict”.
- **pages-site** and **pages-update** (GitHub Pages): 6a65efa “Add GitHub Pages project page”, 765914a “Add content to project page” and 7cd9114 “Add version line to project page and live URL to README”.

Member B: Hafsa El Mahfoudy (Hafsa-mhf)

- **feature-filters:** src/filters.py (filter_by_status, search_tasks) and tests/test_filters.py. Commits: ef8eb52 “Add status filter and keyword search”, 4140218 “Add unit tests for filters” and, after the Request changes review, 7fcc9a8 “Handle non-string keyword in search_tasks”.

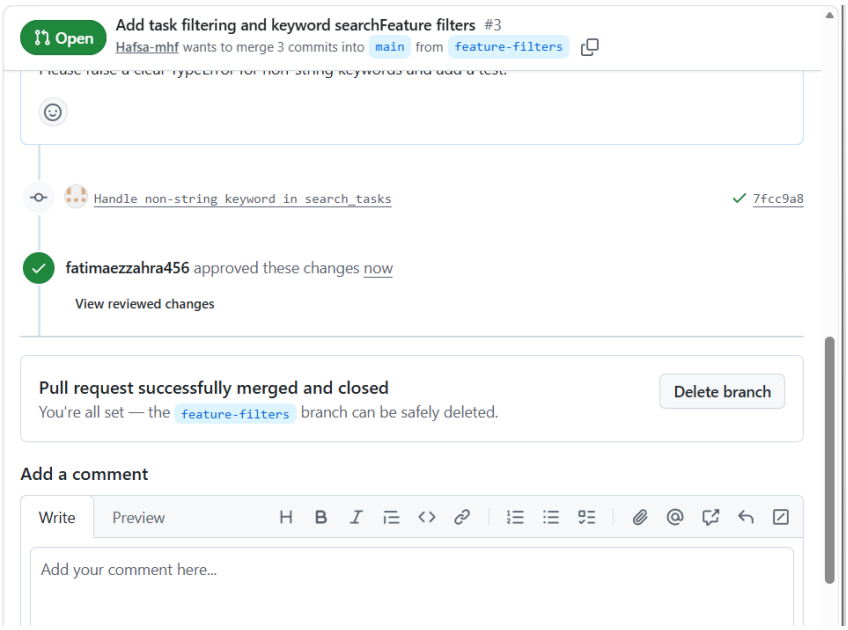
Member C: Chaimae Tahir (chaimaetahir)

- **feature-stats:** src/stats.py (completion_percentage, count_pending) and tests/test_stats.py. Commits: 20324e7 “Add completion percentage and pending count” and f7bd2b9 “Add unit tests for statistics”.
- **status-update-chaima** (merge conflict): 3936188 “Update project status (Chaima)”, changing the README status line to “Core features implemented.”

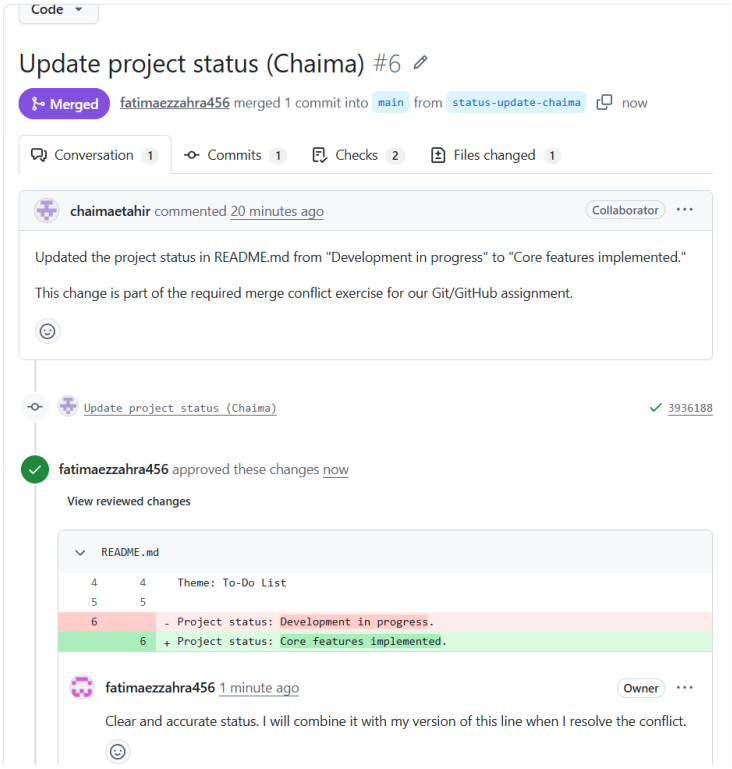
3. Pull Requests & Code Review

- **PR #1, CI workflow** (A): approved by Hafsa-mhf, merged (5780278).
- **PR #2, task management** (A): merged after an approving review.
- **PR #3, filters** (B, Hafsa): reviewed by A with a Request changes review, corrected on the same branch (7fcc9a8), re-reviewed and approved by A, then merged.
- **PR #4, statistics** (C, Chaimae): approved by A, merged (240a87c).
- **PR #5, “Update project status (Fatima)”** (A): the conflict-resolution PR, reviewed by the third member, Hafsa, with a line comment and an approval, then merged.
- **PR #6, “Update project status (Chaima)”** (C): reviewed by A with a line comment on the changed README line and an approval, merged first.
- **PR #7, GitHub Pages project page** (A): approved by Hafsa with a line comment on docs/index.html, merged.
- **PR #8, final page change and live URL in README** (A): approved by Chaimae, merged (a4bb571). Reviews: A reviewed PRs #3, #4 and #6. Hafsa reviewed #1, #5 and #7. Chaimae reviewed #8. Every member authored and reviewed at least one Pull Request.

  Request changes cycle: on PR #3, A asked what happens when search_tasks receives a number such as 123, because it would crash with an AttributeError, and submitted Request changes. Hafsa corrected search_tasks on the **same branch** (feature-filters), committed 7fcc9a8 “Handle non-string keyword in search_tasks” and pushed. The existing PR was reviewed again, A approved it, and it was merged. Because the bypass list is empty, no author could approve or merge their own Pull Request.



PR #3: the requested change, the correction commit 7fcc9a8 on the same branch, then the approval.



PR #6: line comment and approval before the merge.

4. Merge Conflict

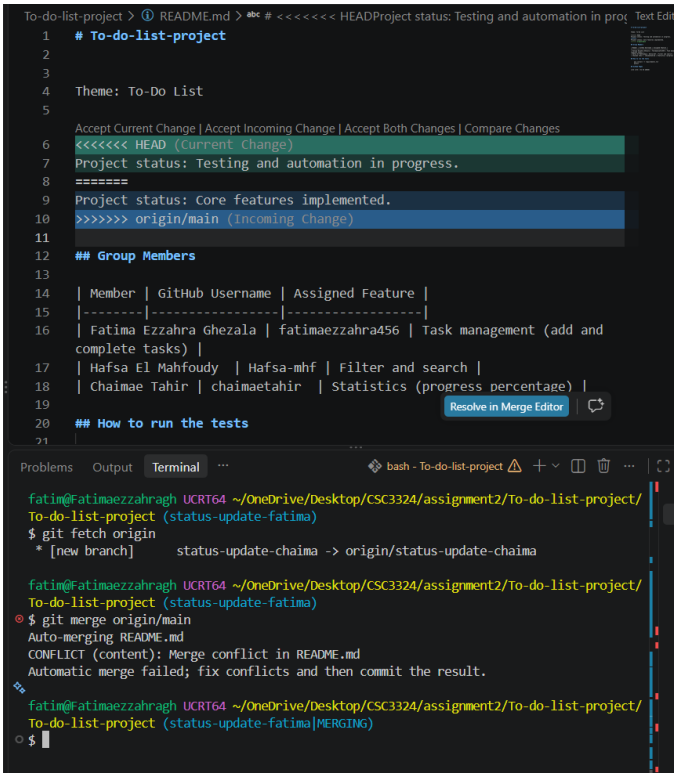
The shared line “Project status: Development in progress.” was added to README.md at the start. Two members then created separate branches from the same version of main and changed that exact line differently:

- **status-update-chaima** (Chaimae, PR #6): “Project status: Core features implemented.” Reviewed by A and merged first.
- **status-update-fatima** (Fatima, PR #5): “Project status: Testing and automation in progress.”

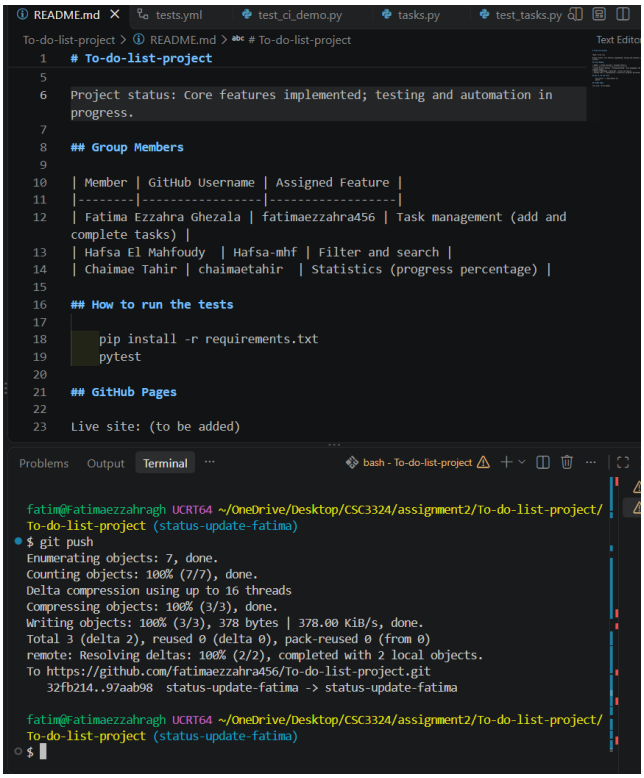
After PR #6 was merged, GitHub showed “This branch has conflicts that must be resolved” on PR #5. On the second branch, A ran git checkout status-update-fatima, git fetch origin and git merge

origin/main. Git reported CONFLICT (content) in README.md. In VS Code, **Current (HEAD)** was A's line ("Testing and automation in progress.") and **Incoming (origin/main)** was Chaimae's line ("Core features implemented.").

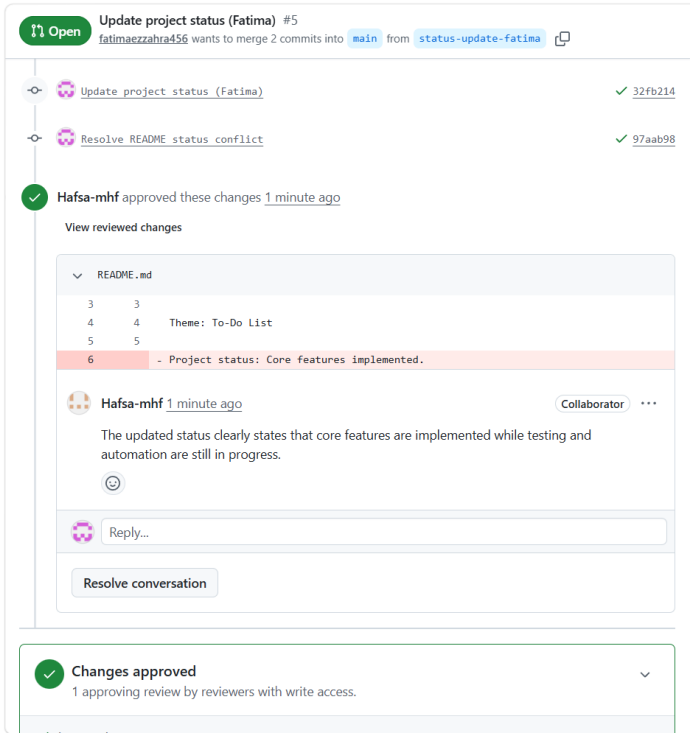
Resolution: both statements were true (the features were implemented and merged, while testing and automation were still being finished), so the group combined them into one line: "Project status: Core features implemented; testing and automation in progress." All conflict markers were removed. A then ran git add README.md, git commit -m "Resolve README status conflict" and git push (32fb214..97aab98). The third member, Hafsa, reviewed PR #5 with a line comment and an approval, and the PR was merged.



Merge conflict in README.md: Current Change, Incoming Change and the CONFLICT message.



Resolved README.md (one combined status line) and the push of the resolution.



PR #5: review by the third member (Hafsa) with a line comment, then “Changes approved”.

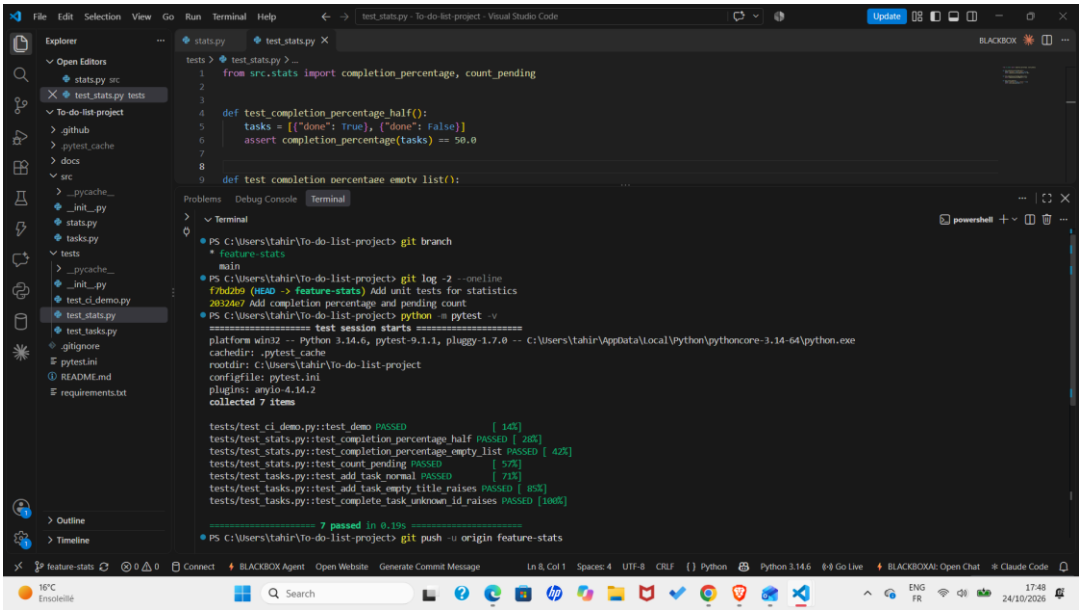
5. Unit Tests

Each member wrote 3 tests with pytest, and the suite also contains the small CI demo test. A run on main (Member C’s machine, before PR #3 was merged) showed **7 passed**: A’s 3 tests, C’s 3 tests and the CI demo test. B’s 3 tests (and the handling of non-string keywords added after the review) came with PR #3.

- **Normal behavior:** test_add_task_normal (add “Buy milk”, check id, title and done flag), test_completion_percentage_half (one of two tasks done gives 50.0),

test_filter_by_status_pending (only the pending task is returned).

- **Edge / boundary cases:** test_completion_percentage_empty_list (an empty list returns 0.0 instead of dividing by zero) and test_search_empty_keyword_returns_all (an empty keyword returns every task).
- **Invalid input / error handling:** test_add_task_empty_title_raises (a whitespace-only title raises ValueError) and test_complete_task_unknown_id_raises (an unknown id raises KeyError). Hafsa's correction rejects non-string keywords such as 123 with a TypeError.



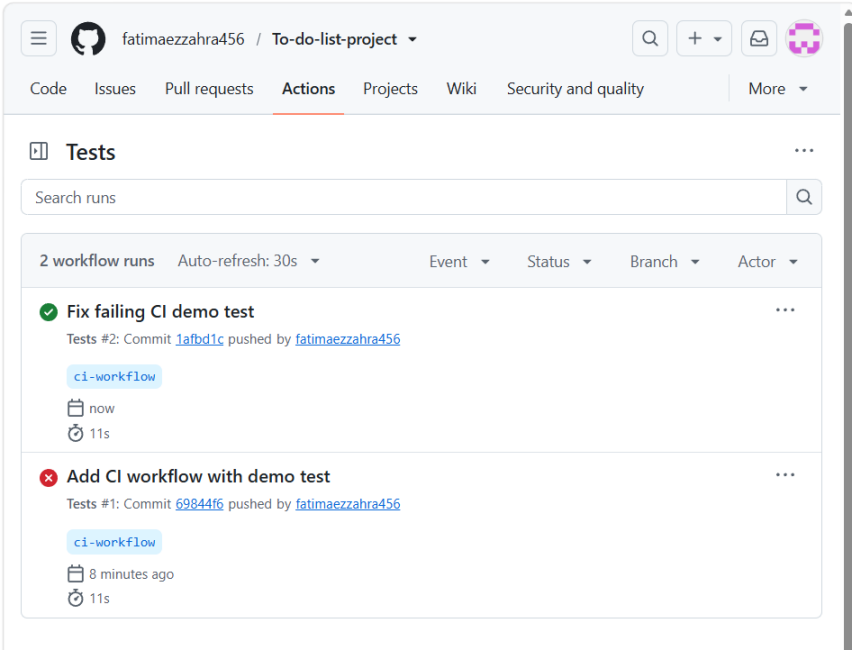
Local pytest run: 7 passed (tests of Member A and Member C and the CI demo test).

6. Continuous Integration

The workflow .github/workflows/tests.yml (name “Tests”) runs on every **push** and every **pull_request**. It has one job, **test**, on an **ubuntu-latest** runner with these steps: checkout the code (actions/checkout@v4), set up Python 3.12 (actions/setup-python@v5), install dependencies (pip install -r requirements.txt, which installs pytest) and run the whole suite with pytest.

Failed run: Tests #1 (commit 69844f6, “Add CI workflow with demo test”) failed. The cause was the deliberately wrong assertion in tests/test_ci_demo.py (assert 1 + 1 == 3). The log shows that checkout, Python setup and dependency installation passed, and only the Run tests step failed with AssertionError, “1 failed” and exit code 1.

Fix: the group changed the assertion to 1 + 1 == 2, committed it on the same branch (1afbd1c, “Fix failing CI demo test”) and pushed. Tests #2 then finished with a green check. All later Pull Requests (#3 to #8) were checked automatically by the same workflow.



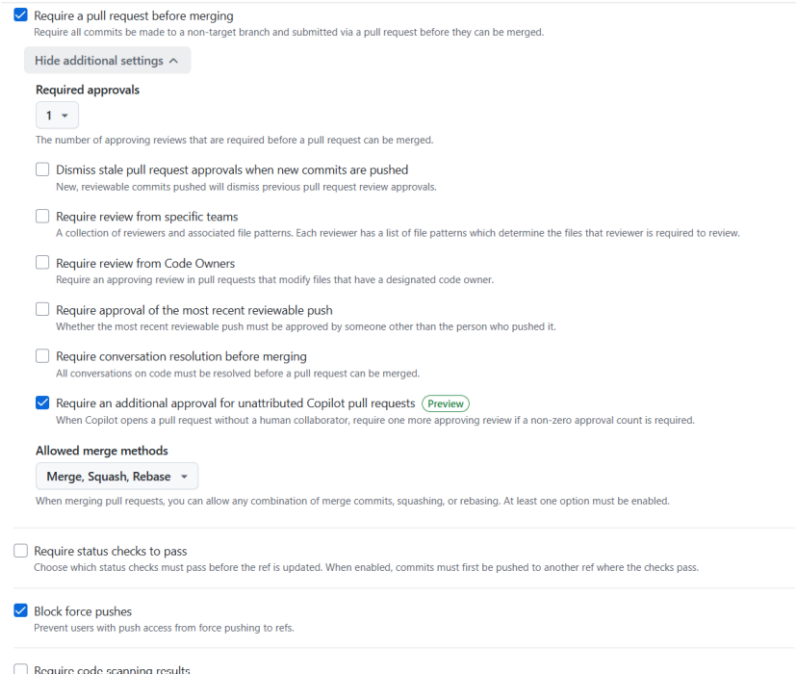
Actions tab: Tests #2 (green) after Tests #1 (red).

7. Ruleset & Protected main

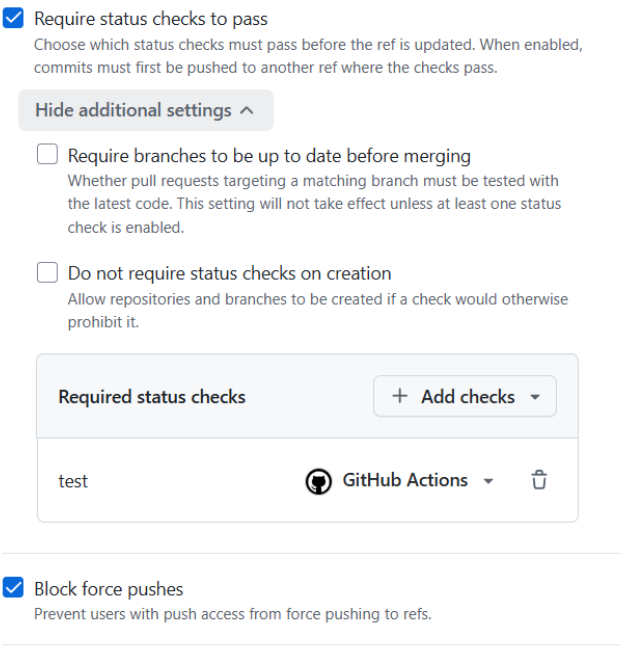
The ruleset **protect-main** has enforcement **Active** and targets the default branch, main. The **bypass list is empty**. Its rules are:

- Restrict deletions.
- Require a pull request before merging, with **1 required approval**.
- Block force pushes.
- Require status checks to pass, with the required check **test** (GitHub Actions), added after the workflow had run.

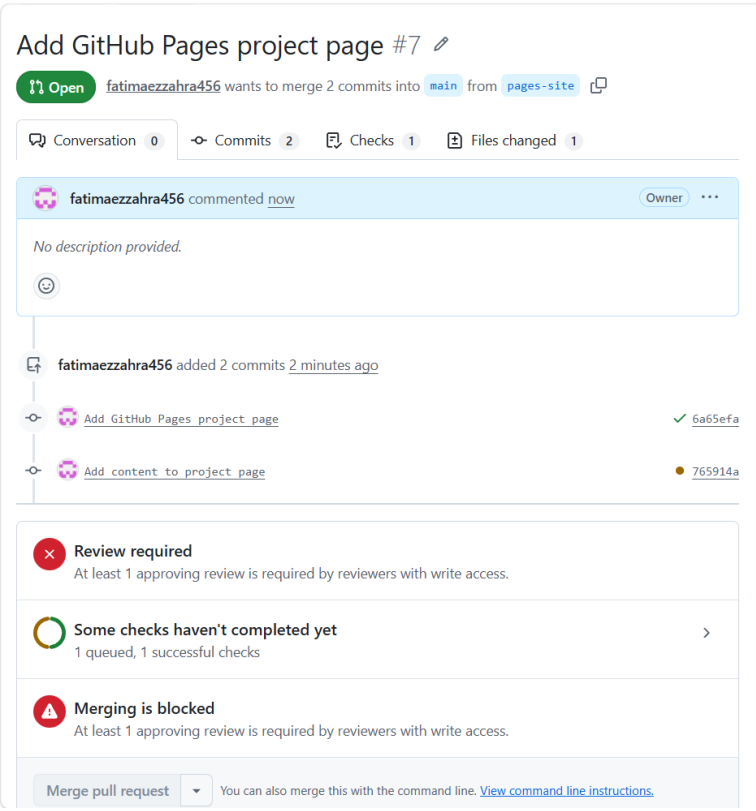
A Pull Request therefore needed both an approval from a member other than the author and the successful automated check before it could be merged. PR #7 shows this: the merge button was blocked by “Review required” while the test check was still queued. It was merged only after Hafsa approved it and the checks passed. The same happened for PRs #1 to #6 and #8.



Required approvals: 1, and Block force pushes enabled.



Required status check: test (GitHub Actions).



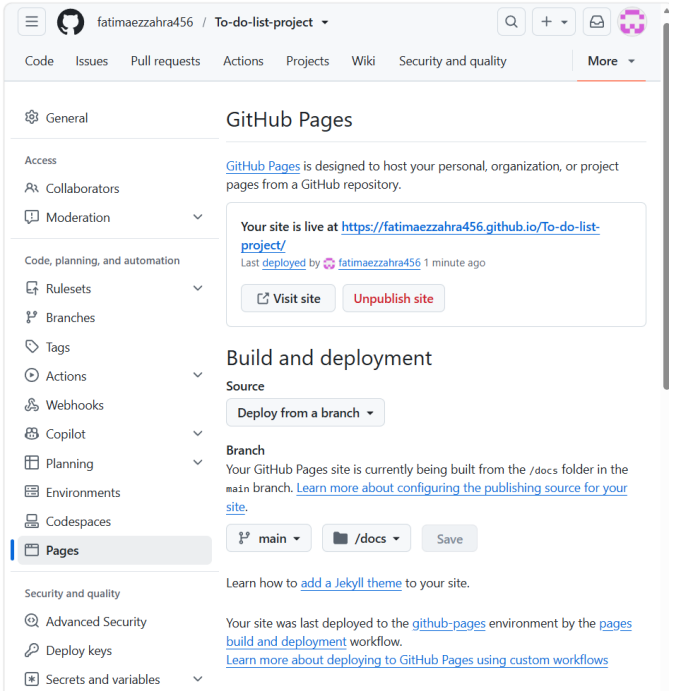
PR #7: merging is blocked by “Review required” while the check is queued.

8. Continuous Deployment

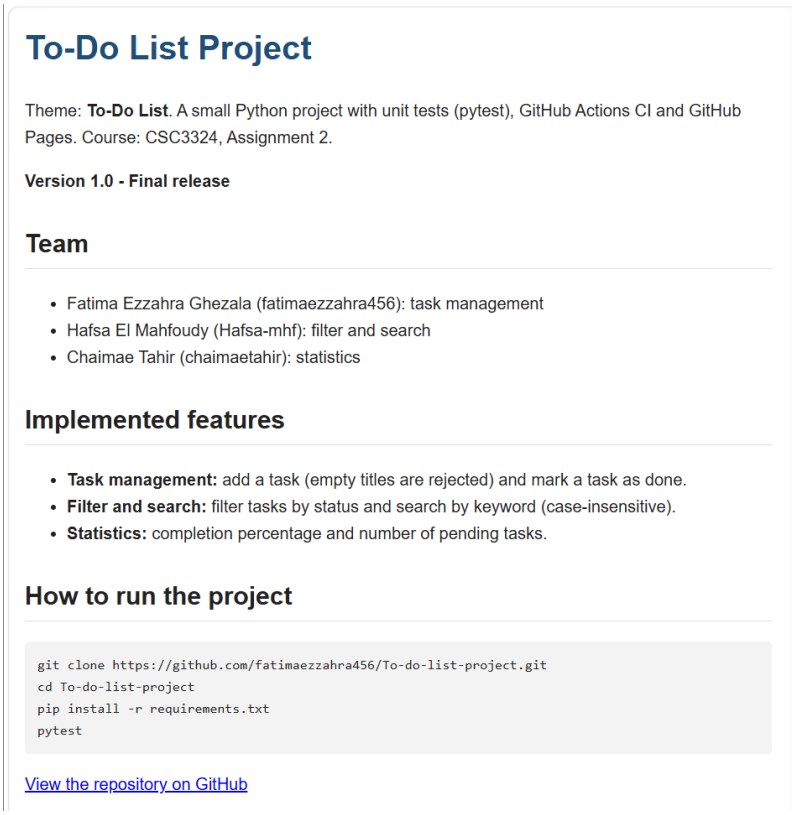
The site is published with **GitHub Pages** from the **main** branch and the **/docs** folder (Settings, Pages, Deploy from a branch). Every merge into main that changes docs/ is redeployed automatically by the “pages build and deployment” workflow. Live URL: https://fatimaezzahra456.github.io/To-do-list-project/

The page docs/index.html (branch pages-site, PR #7) contains the project title, the theme, the team members, the implemented features and the instructions to run the project.

Final change: on the branch pages-update (commit 7cd9114, PR #8), the line “Version 1.0 - Final release” was added to the page and the live URL was added to README.md. After the review by Chaimae and the merge (a4bb571), the live site showed the new line without any manual deployment.



GitHub Pages settings: published from main and /docs, with the live URL.



To-Do List Project

Theme: **To-Do List**. A small Python project with unit tests (pytest), GitHub Actions CI and GitHub Pages. Course: CSC3324, Assignment 2.

Version 1.0 - Final release

Team

- Fatima Ezzahra Ghezala (fatimaezzahra456): task management
- Hafsa El Mahfoudy (Hafsa-mhf): filter and search
- Chaimae Tahir (chaimaetahir): statistics

Implemented features

- **Task management:** add a task (empty titles are rejected) and mark a task as done.
- **Filter and search:** filter tasks by status and search by keyword (case-insensitive).
- **Statistics:** completion percentage and number of pending tasks.

How to run the project

```
git clone https://github.com/fatimaezzahra456/To-do-list-project.git
cd To-do-list-project
pip install -r requirements.txt
pytest
```

View the repository on GitHub

Live site after PR #8: the line “Version 1.0 - Final release” is visible.

9. Contribution Summary

Fatima Ezzahra Ghezala (fatimaezzahra456), Member A

- Feature: task management. Branches: feature-tasks, ci-workflow, status-update-fatima, pages-site, pages-update.
- PRs authored: #1, #2, #5, #7, #8. PRs reviewed: #3 (Request changes, then approval), #4, #6.
- Tests: 3 (tests/test_tasks.py). Also the ruleset, the CI workflow and its intentional failure, and the Pages site.

Hafsa El Mahfoudy (Hafsa-mhf), Member B

- Feature: filter and search. Branch: feature-filters.
- PR authored: #3. PRs reviewed: #1, #5, #7.
- Tests: 3 (tests/test_filters.py) and the correction for non-string keywords.

Chaimae Tahir (chaimaetahir), Member C

- Feature: statistics. Branches: feature-stats, status-update-chaima.
- PRs authored: #4, #6. PR reviewed: #8.
- Tests: 3 (tests/test_stats.py).

10. Fair Use of AI

Our group used AI as a support tool to help with coding, testing, troubleshooting, and understanding GitHub workflows. We reviewed the suggestions and adapted them to our project requirements. AI was used to assist our work, while the group remained responsible for the final results.

11. Sources & Reflection

Sources: GitHub Documentation, Python Documentation, and AI tools used during the project.

One of the main challenges was coordinating our work and making sure that everyone's contributions worked together correctly. We learned the importance of testing changes and reviewing code before merging it. If a Pull Request, merge, test, or workflow behaves unexpectedly, we would first check the error messages, review the recent changes, and verify that the code and configuration are correct